Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Explain the importance of quality considerations during the requirement elicitation phase.	**2**	(CO1, L2)
2)	Explain the significance of understanding different layers in software architecture.	**2**	(CO2, L2)
3)	Examine the differences between Data Definition Language (DDL) and Data Manipulation Language (DML) queries in database management.	**2**	(CO3, L4)
4)	Examine the concept of a relational model in database design by explaining its key components and analyzing how these components contribute to data integrity. Provide one example to illustrate your explanation.	**2**	(CO1, L4)
5)	Determine the quality attributes that should be considered during the requirement elicitation phase to ensure effective software development.	**2**	(CO2, L3)

Section B **5*4=20**

6)	Discuss the importance of different database layers and provide an example of how DDL and DML queries can be used to manage data within these layers.	**5**	(CO3, L2)
7)	Analyze the concept of relational views in database management systems by explaining their purpose, advantages, and disadvantages compared to base tables. Provide examples to support your answer.	**5**	(CO1, L4)
8) a)	Analyze the challenges faced during the requirement elicitation phase and discuss how these challenges can impact the quality of the final software product. Provide an example to illustrate your points.	**5**	(CO2, L4)
	OR		
b)	Examine the role of views in database management systems by analyzing their advantages and disadvantages. Use examples to differentiate between materialized and non-materialized views.	**5**	(CO2, L4)

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9)	Construct a simple database schema for an online bookstore using DDL statements and explain how you would update the price of a specific book using a DML query.	**5**	(CO3, L3)

Section C **10*2=20**

10)	Determine and construct an ethical framework for developing a software product that addresses user privacy concerns during the requirement elicitation phase. Your answer should include steps to ensure the quality of the software while respecting potential legal and ethical considerations.	**10**	(CO1, L3)
11) a)	Examine the relational model by analyzing its structure using entity types and relationships. Further, differentiate between a base table (relation schema) and a view in terms of their representation and usage. Finally, compare and contrast the advantages and limitations of using views in database management systems.	**10**	(CO2, L4)
	OR		
b)	Analyze and differentiate between the conceptual, logical, and physical layers of database management systems. For each layer, explain its significance and how it interacts with other layers.	**10**	(CO2, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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